Theory of Automata Assignment

Date of Submission: 30 June

Q1. Construct a Regular Expression and DFA for strings over {a,b} that end with 'ab'.

Q2. Construct a Regular Expression and DFA for strings containing an even number of a's.

Q3. Construct a Regular Expression and DFA for strings containing an odd number of b's.

Q4. Construct a DFA that accepts all strings that start with 'a'. Also write the corresponding Regular Expression.

Q5. Construct a DFA and Regular Expression for strings that contain the substring 'aa'.

Q6. Design a DFA for binary strings ending with '01'. Write its Regular Expression as well.

Q7. Construct a DFA that accepts all strings whose length is a multiple of 3 over Σ={a,b}.

Q8. Design a DFA for strings that start and end with the same symbol. Also write the corresponding Regular Expression.

Q9. Construct a DFA that accepts all strings containing the substring '101'. Write the corresponding Regular Expression.

Q10. Write Regular Expressions for:

(a) all strings over {a,b}

(b) strings starting with 'ab'

(c) strings ending with 'ba'

(d) strings containing exactly one 'a'

(e) strings containing at least one 'b'.

Q11. Convert the following Regular Expressions into DFA: (a+b)*ab, aa(a+b)*, (0+1)*101.

Q12. Write Regular Expressions for: even number of a's, odd number of b's, strings ending with aa, strings starting with bb, binary strings ending with 11.

Q13. Construct a DFA and Regular Expression for strings containing exactly two a's.

Q14. Construct a DFA for binary strings where the number of 1's is divisible by 3. Explain each state.

Q15. Design a DFA in which every 'a' is immediately followed by a 'b'.

Q16. Construct a DFA for strings that do not contain the substring 'aaa'.

Q17. Construct a DFA and Regular Expression for strings containing both substrings 'ab' and 'ba'.

Q18. Design a DFA that accepts binary strings having an even number of 0's and an odd number of 1's.

Q19. Construct a DFA for strings whose third symbol from the end is 'a'.

Q20. Construct a DFA for strings that start with 'a' and contain an even number of b's.

Q21. Write Regular Expressions for: exactly one occurrence of 'ab', at least two consecutive a's, strings not containing 'bb', strings ending with 'aba', strings having odd length.

Q22. Convert the following Regular Expressions into DFA: (a+b)*abb, a*b(a+b)*, (0+1)*1011.

Q23. Design a DFA for strings where length(w) mod 4 = 0.

Q24. Construct a DFA where the number of a's and b's are both even.

Q25. Construct a DFA for strings that contain neither 'aa' nor 'bb'.